

| Layer (type:depth-idx)         | Input Shape   | Output Shape  | Param # |
|--------------------------------|---------------|---------------|---------|
| TransformerASR                 | [16, 100, 80] | [16, 50, 400] | --      |
| └Sequential: 1-1               | [16, 80, 100] | [16, 192, 25] | --      |
| └Conv1d: 2-1                   | [16, 80, 100] | [16, 192, 50] | 46,272  |
| └GELU: 2-2                     | [16, 192, 50] | [16, 192, 50] | --      |
| └Conv1d: 2-3                   | [16, 192, 50] | [16, 192, 25] | 110,784 |
| └GELU: 2-4                     | [16, 192, 25] | [16, 192, 25] | --      |
| └Linear: 1-2                   | [16, 25, 192] | [16, 25, 192] | 37,056  |
| └PositionalEncoding: 1-3       | [16, 25, 192] | [16, 25, 192] | --      |
| └Dropout: 2-5                  | [16, 25, 192] | [16, 25, 192] | --      |
| └TransformerEncoder: 1-4       | [16, 25, 192] | [16, 25, 192] | --      |
| └ModuleList: 2-6               | --            | --            | --      |
| └TransformerEncoderLayer: 3-1  | [16, 25, 192] | [16, 25, 192] | 247,744 |
| └TransformerEncoderLayer: 3-2  | [16, 25, 192] | [16, 25, 192] | 247,744 |
| └TransformerEncoderLayer: 3-3  | [16, 25, 192] | [16, 25, 192] | 247,744 |
| └TransformerEncoderLayer: 3-4  | [16, 25, 192] | [16, 25, 192] | 247,744 |
| └TransformerEncoderLayer: 3-5  | [16, 25, 192] | [16, 25, 192] | 247,744 |
| └TransformerEncoderLayer: 3-6  | [16, 25, 192] | [16, 25, 192] | 247,744 |
| └Embedding: 1-5                | [16, 50]      | [16, 50, 192] | 76,800  |
| └PositionalEncoding: 1-6       | [16, 50, 192] | [16, 50, 192] | --      |
| └Dropout: 2-7                  | [16, 50, 192] | [16, 50, 192] | --      |
| └TransformerDecoder: 1-7       | [16, 50, 192] | [16, 50, 192] | --      |
| └ModuleList: 2-8               | --            | --            | --      |
| └TransformerDecoderLayer: 3-7  | [16, 50, 192] | [16, 50, 192] | 396,352 |
| └TransformerDecoderLayer: 3-8  | [16, 50, 192] | [16, 50, 192] | 396,352 |
| └TransformerDecoderLayer: 3-9  | [16, 50, 192] | [16, 50, 192] | 396,352 |
| └TransformerDecoderLayer: 3-10 | [16, 50, 192] | [16, 50, 192] | 396,352 |
| └TransformerDecoderLayer: 3-11 | [16, 50, 192] | [16, 50, 192] | 396,352 |
| └TransformerDecoderLayer: 3-12 | [16, 50, 192] | [16, 50, 192] | 396,352 |
| └Linear: 1-8                   | [16, 50, 192] | [16, 50, 400] | 77,200  |

Total params: 4,212,688  
 Trainable params: 4,212,688  
 Non-trainable params: 0  
 Total mult-adds (M): 103.53

Input size (MB): 0.52  
 Forward/backward pass size (MB): 61.54  
 Params size (MB): 6.18  
 Estimated Total Size (MB): 68.24